PYTHON FOR SUPPLY CHAIN

Session 5 · Mon 30 Nov 2026

Star products

groupby, running totals, ABC classes, a cold test



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Today

1

Load and check

`(101458, 10)`

2

Group and sum

`groupby` `sum`

3

ABC classes

`cumsum` `apply`

4

Cold test

`20 minutes, notes allowed`

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Start from your clean table

Session 4

`sales_clean.csv`



Session 5

starts here

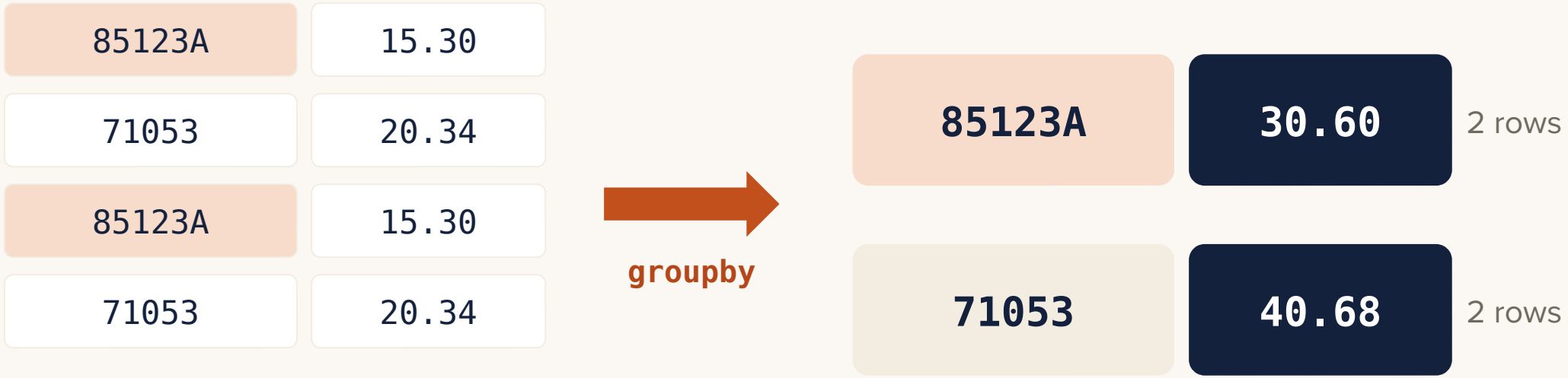
```
sales_clean["invoice_date"] = pd.to_datetime(sales_clean["invoice_date"])
sales_clean.shape
```

```
→ (101458, 10)
```

A wrong shape: stop and fix it first.

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groupby sorts receipts into piles



Like a PivotTable: rows = stock_code, values = sum of revenue

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Read a groupby from left to right

```
sales_clean .groupby("stock_code") ["revenue"] .sum()
```

the table — the piles — the column — add each pile

`.sum()` total

`.mean()` average

`.count()` how many rows

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reset_index makes a normal table

Series

22423	51504.85
85123A	30780.98
79321	17013.92



reset_index()

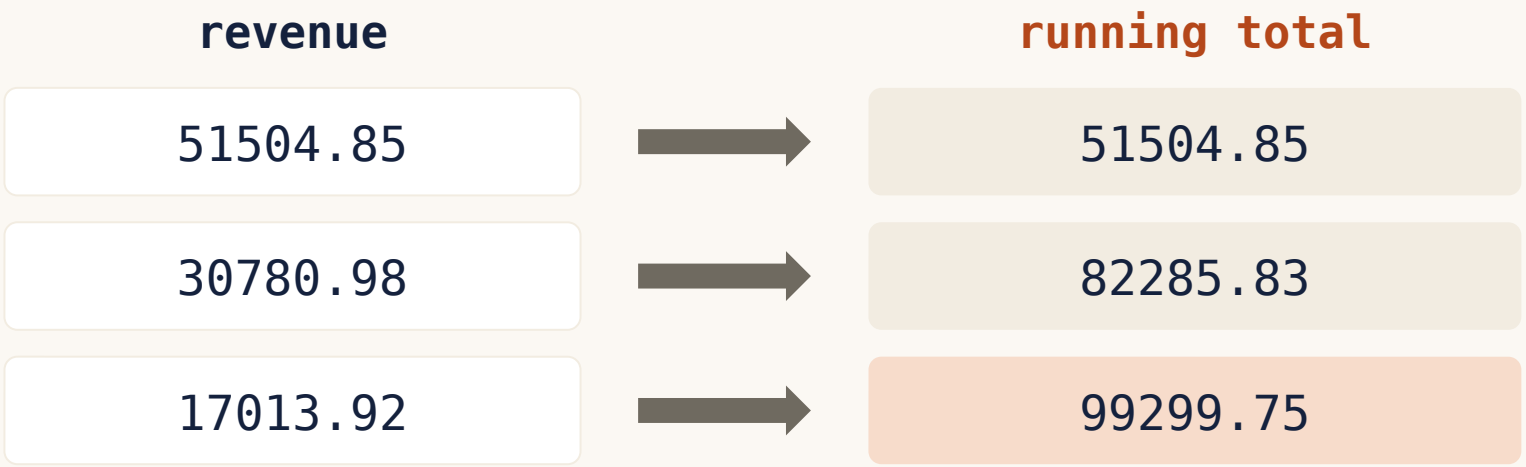
a normal table

stock_code	revenue
22423	51504.85
85123A	30780.98
79321	17013.92

```
product_value = revenue_by_product.reset_index()
```

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cumsum is a running total

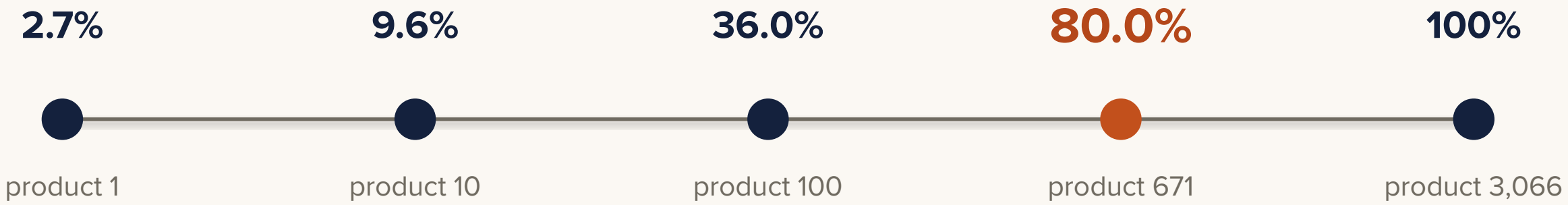


Sort first, then cumsum.

Like =SUM(B2:B2) dragged down

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Cumulative share: how far we are




```
product_value["cumulative_share"] = product_value["revenue"].cumsum() / total_revenue
```

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ABC: three classes of product

A

up to 80%

watch closely

B

80% to 95%

check often

C

the rest

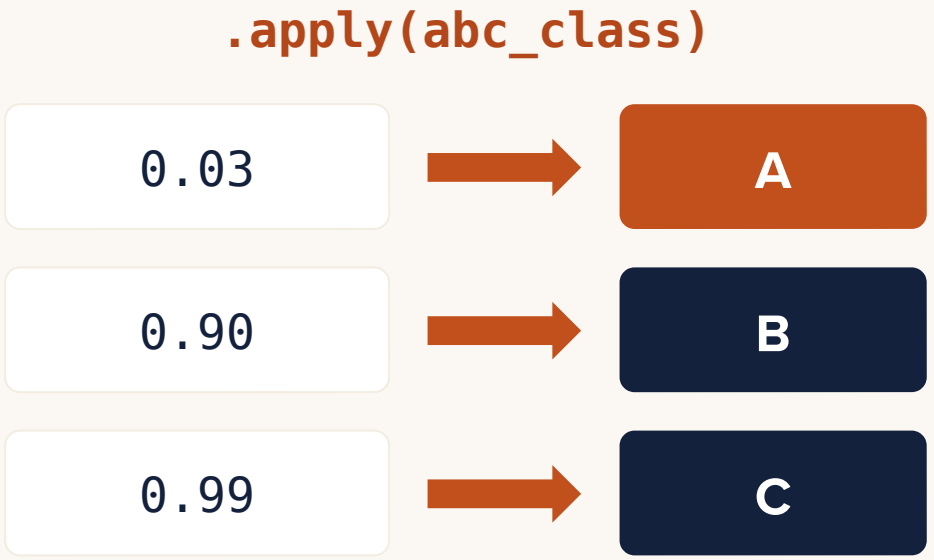
keep simple

80 and 95 are a habit, not a law.

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apply runs your rule on every row

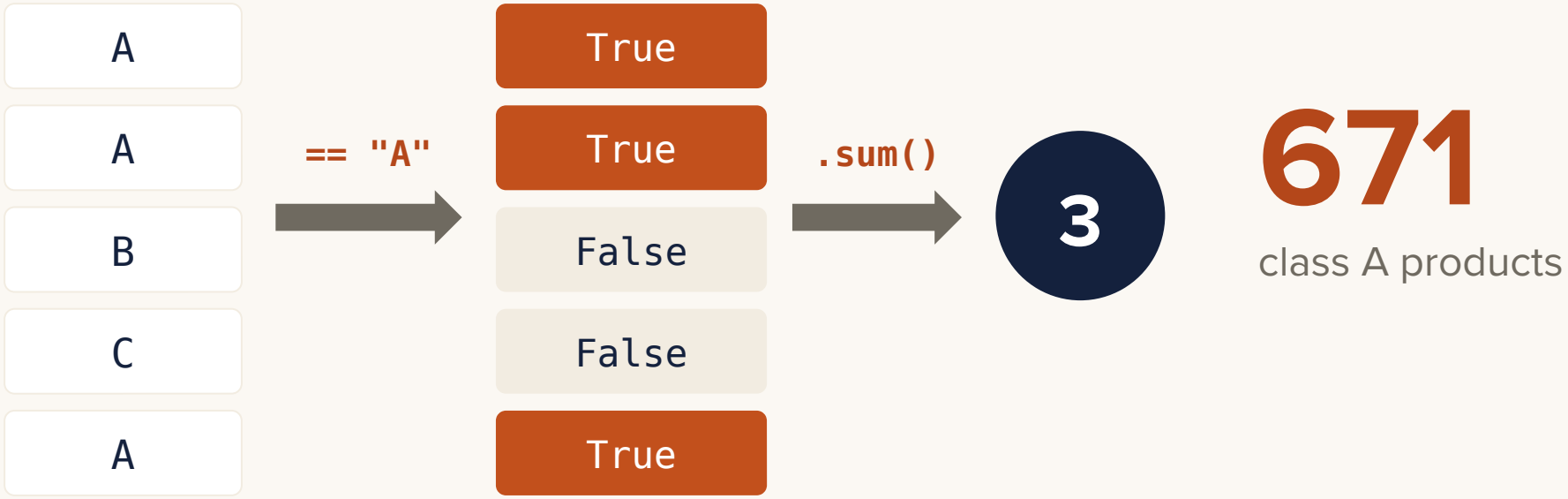
```
def abc_class(share):
    if share <= 0.80:
        return "A"
    elif share <= 0.95:
        return "B"
    else:
        return "C"
```



Your Session 3 rule, on every row

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Count with True and sum



671

class A products

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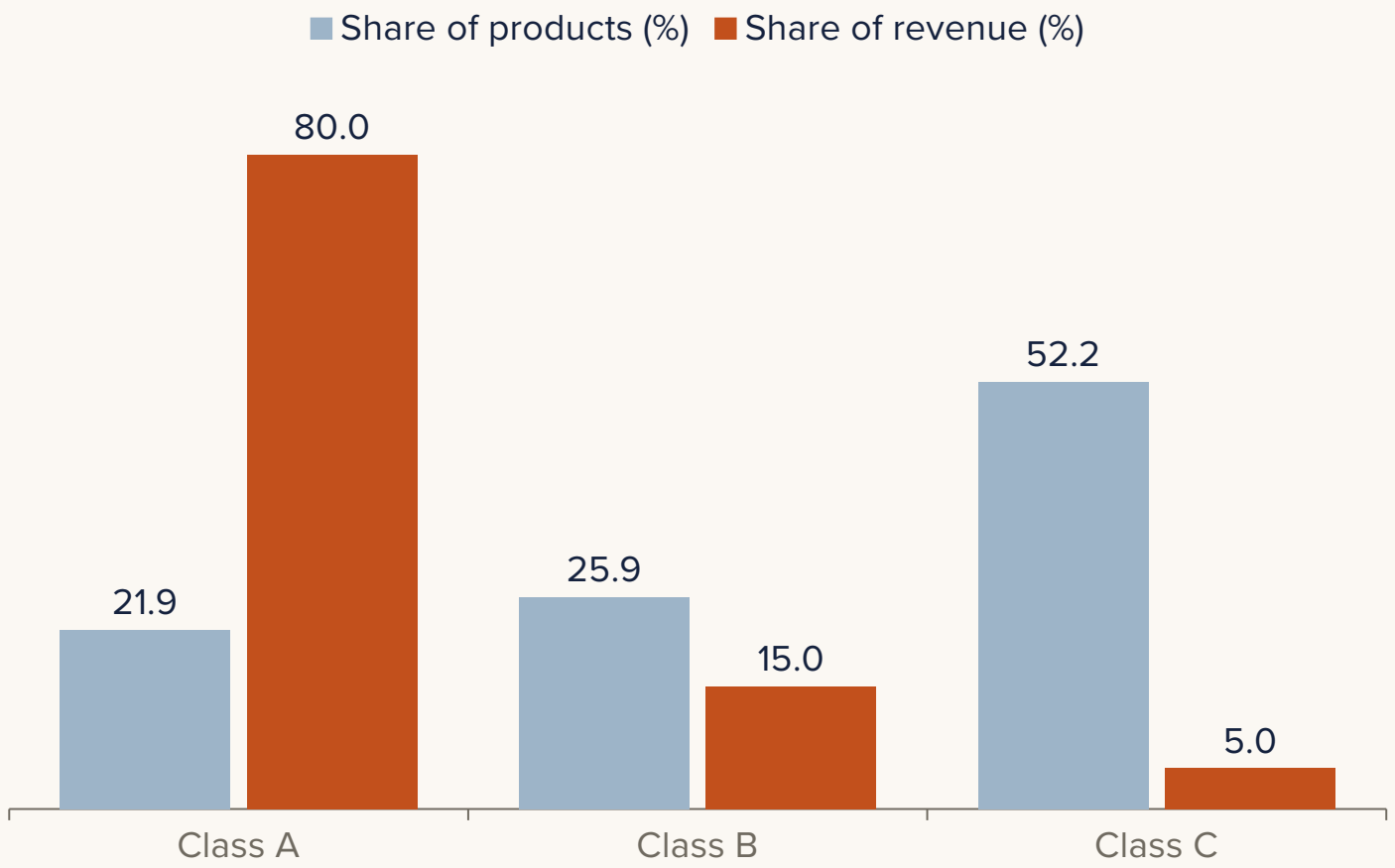
About 1 in 5 products bring 80% of the money

21.9%

of products

80.0%

of revenue



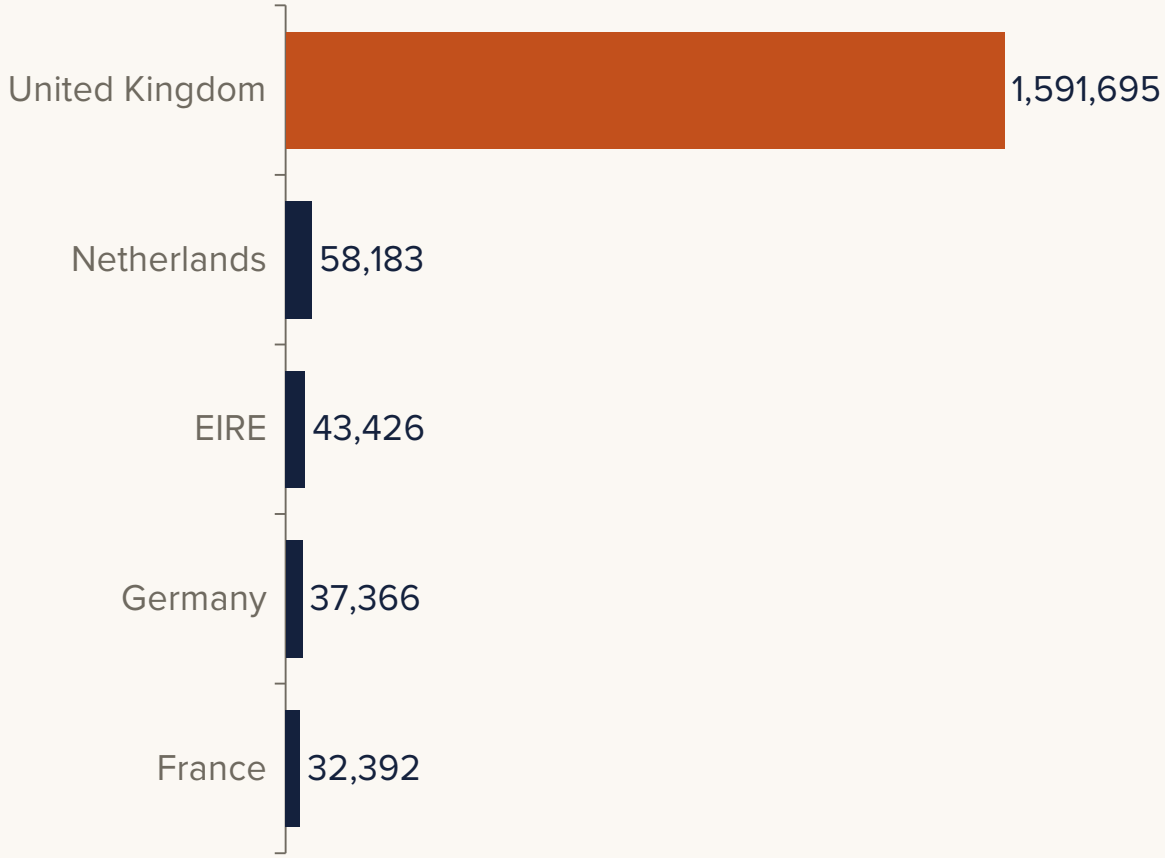
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Same groupby, another pile

```
groups = sales_clean.groupby("country")
groups["revenue"].sum()
```

UK is 84.8% of revenue

Country charts are not global.



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Cold test: 20 minutes

1

It runs

top to bottom

1

Right numbers

within 0.01

1

Clear names

says what it holds

1

One sentence

what it means

Notes allowed. The teacher stays silent.

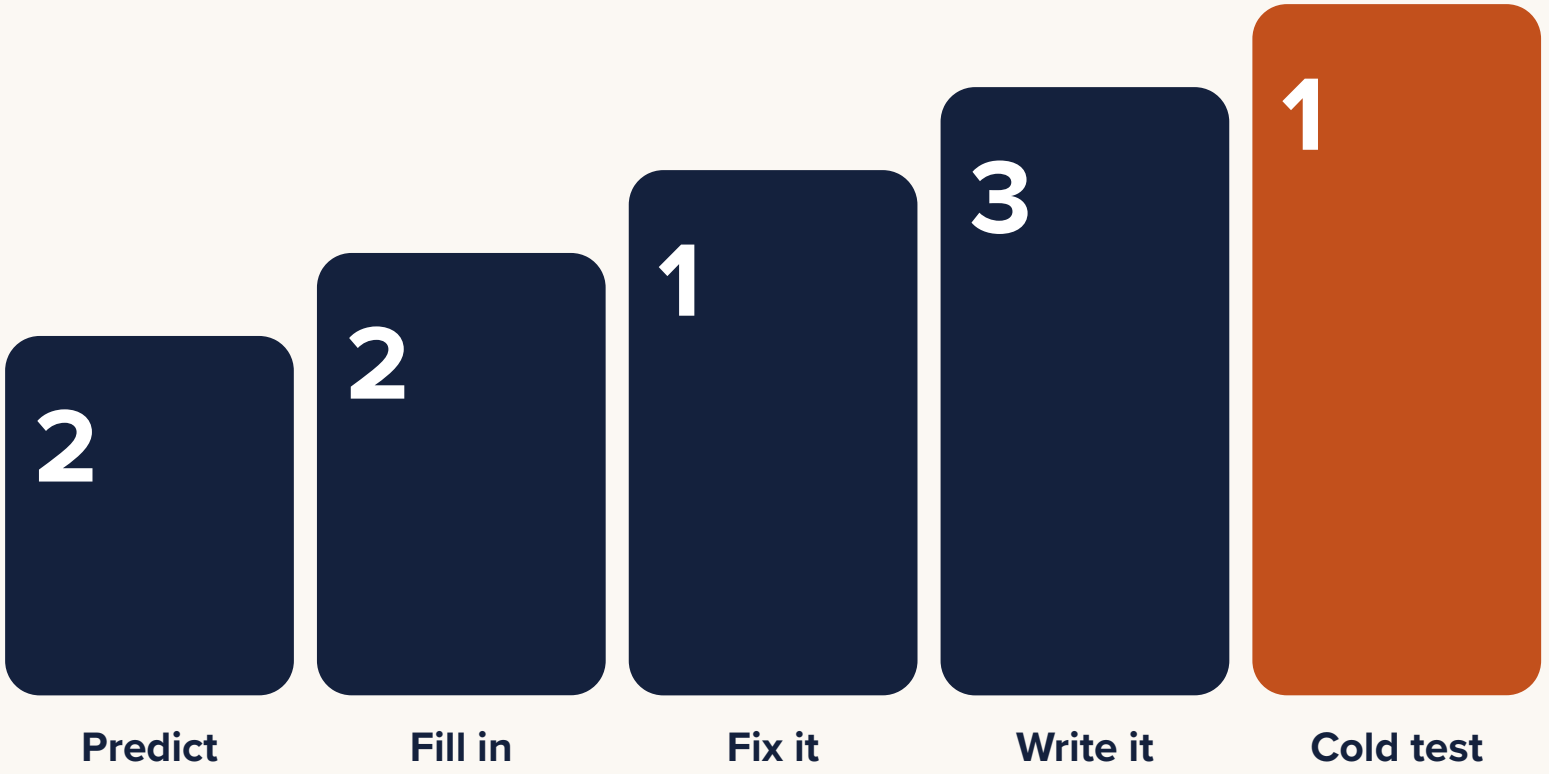
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Your analysis toolkit

Code	Does
`df.groupby("col")["revenue"].sum()`	total per group
`.sort_values(ascending=False)`	biggest first
`.reset_index()`	back to a normal table
`.cumsum()`	running total
`.apply(abc_class)`	your rule on every row
`(df["abc_class"] == "A").sum()`	count the True rows
`.to_csv(path, index=False)`	save for later

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Now in Colab



Notebook

s5_star_products

Save

product_value.csv

Recall

Tue 1, Wed 2 Dec

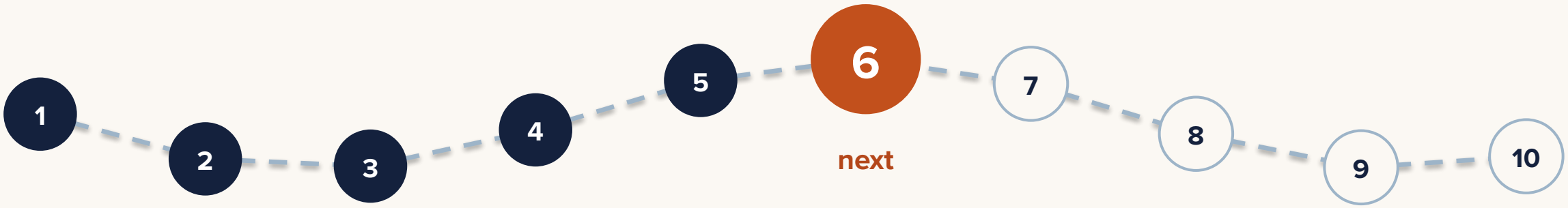
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Thank you

Next: Session 6

How do the numbers talk?

Thu 3 Dec



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